Unit 7 Bellwork – Week 2 Name: ______________________ Period: _____

Day 1: Synthetic Division

1. Use synthetic division: $(x^3 + 5x^2 + 7x + 3) \div (x + 1)$

$$\begin{array}{r|rrrr} -1 & 1 & 5 & 7 & 3 \\ \hline & & & & \end{array}$$

Quotient: ____________________

2. Is $x = 2$ a root of $f(x) = x^3 - 6x^2 + 11x - 6$? Use synthetic division.

$$\begin{array}{r|rrrr} 2 & 1 & -6 & 11 & -6 \\ \hline & & & & \end{array}$$

Remainder: ________ Root? YES / NO

Day 2: Box Method Division

1. Use the box method: $(x^3 + 2x^2 - 5x - 6) \div (x + 1)$

	x^2		
x	x^3		
$+1$			

Quotient: ____________________

2. **Spiral:** Simplify $\sqrt{72x^4} =$ ____________________

Day 3: Sum & Difference of Cubes

Formulas: $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$ $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$ **SOAP:** Same, Opposite, Always Positive

1. Factor: $x^3 + 8 = (x +$ ____$)(x^2 -$ ____ $+$ ____$)$

2. Factor: $x^3 - 27 =$ ___________________________

3. Factor: $8x^3 + 125 =$ ___________________________

4. **SAT:** Which is a factor of $x^3 - 64$?

A. $(x - 8)$ B. $(x + 4)$ C. $(x - 4)$ D. $(x^2 + 4)$

Day 4: Factoring by Grouping

1. Factor: $x^3 + 2x^2 + 5x + 10$

Group: $(x^3 + 2x^2) + (5x + 10) = x^2($________$) + 5($________$) =$ ____________________

2. Factor completely: $x^3 - 3x^2 - 4x + 12$ (Hint: Check if you can factor further!)

Answer: _________________________________

3. **Spiral:** Simplify: $(2x^{-3})^2 =$ _____________

Day 5: Mixed Review + SAT/TSI Practice

1. Factor: $2x^3 + 16$ (GCF first!) $=$ ________________________

2. **SAT:** What is the value of $27^{2/3}$?

   A. 3 B. 9 C. 18 D. 729

3. **TSI:** Which expression equals $(x-2)(x^2+2x+4)$?

   A. x^3-8 B. x^3+8 C. x^3-4 D. x^3-2

4. **SAT:** If $f(x) = x^3 - 4x^2 + x + 6$ and $f(2) = 0$, which is a factor of $f(x)$?

   A. $(x+2)$ B. $(x-2)$ C. $(x+6)$ D. $(x-6)$

Answer Key: D1: 1. x^2+4x+3 2. R=0, Yes **D2:** 1. x^2+x-6 2. $6x^2\sqrt{2}$ **D3:** 1. $(x+2)(x^2-2x+4)$ 2. $(x-3)(x^2+3x+9)$ 3. $(2x+5)(4x^2-10x+25)$ 4. C **D4:** 1. $(x+2)(x^2+5)$ 2. $(x-3)(x-2)(x+2)$ 3. $4/x^6$ **D5:** 1. $2(x+2)(x^2-2x+4)$ 2. B 3. A 4. B